EXPERIMENT-7

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341 lines (341 loc) · 6.43 KB

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Experiment 7: J Functions

1. Write a Python function to find the maximum and minimum numbers from a sequence of numbers. (Note: Do not use built-in functions.)

In [ ]:

```
def find_max_min(numbers):
    if len(numbers) == 0:
        return None

    max_num = numbers[0]
    min_num = numbers[0]

    for num in numbers:
        if num > max_num:
            max_num = num
        if num < min_num:
            min_num = num

    return max_num, min_num


nums = [10, 6, 8, 90, 12, 56]
print(find_max_min(nums))
```

```
(90, 6)
```

2. Write a Python function that takes a positive integer and returns the sum of the cube of all the positive integers smaller than the specified number.

In [ ]:

```
def sum_of_cubes(n):
    total = 0
    for i in range(1, n):
        total += i ** 3
    return total



print(sum_of_cubes(5))
```

```
100
```

3. Write a Python function to print 1 to n using recursion. (Note: Do not use loop)

In [ ]:

```
def print_numbers(n):
    if n == 0:
        return
    print_numbers(n - 1)
    print(n)



print_numbers(5)
```

```
1
2
```

```
3
4
5
```

4. Write a recursive function to print Fibonacci series upto n terms.

In [ ]:

```python
def fibonacci(n):
    if n <= 1:
        return n
    return fibonacci(n - 1) + fibonacci(n - 2)


def print_fibonacci(n):
    for i in range(n):
        print(fibonacci(i), end=" ")



print_fibonacci(7)
```

```
0 1 1 2 3 5 8
```

5. Write a lambda function to find volume of cone.

In [ ]:

```python
import math

volume_cone = lambda r, h: (1/3) * math.pi * r**2 * h


print(volume_cone(3, 5))
```

```
47.12388980384689
```

6. Write a lambda function which gives tuple of max and min from a list.
   Sample input: [10, 6, 8, 90, 12, 56]
   Sample output: (90,6)

In [ ]:

```python
max_min = lambda lst: (
    max(lst), min(lst)
)

# Sample input
data = [10, 6, 8, 90, 12, 56]
print(max_min(data))
```

```
(90, 6)
```

7. Write functions to explain mentioned concepts:
   a. Keyword argument
   b. Default argument
   c. Variable length argument

In [ ]:

```python
#a)
def student(name, age):
    print("Name:", name)
    print("Age:", age)
```

```
student(age=20, name="Krish")
```

```
Name: Krish
Age: 20
```

In [1]:

```
#b)
def greet(name="Guest"):
    print("Hello", name)

greet("Shreyansh")
greet()
```

```
Hello Shreyansh
Hello Guest
```

In [ ]:

```
#c)
def total_marks(*marks):
    total = 0
    for m in marks:
        total += m
    print("Total:", total)

total_marks(10, 20, 30)
```

```
Total: 60
```